

Element B

Attachment System:

After assessing the problem of safely containing and transporting Komodo dragons, we began investigating existing solutions for similar species. Unfortunately, there is limited published information specific to Komodo dragon containment. Because Komodo dragons are still relatively new additions to many zoo collections, standard caging systems for them have not yet been fully developed. However, comparable reptiles such as crocodiles and alligators present similar containment challenges and offered a valuable starting point for our research.



(Photo courtesy of Kathleen Calderwood, ABC News)

The most common crocodile/alligator cage design uses a long, segmented cage that is also collapsible. The purpose of our cage design is to secure the animal while allowing zookeepers to safely perform medical procedures, such as injections and x-rays, without coming into direct contact with the animal. Most often, the cages are extremely long to ensure that any size animal fits inside without restriction.

Through careful examination of these systems, we found that segmented and collapsible cage designs are the most commonly used for large reptiles. These designs are typically constructed from strong materials like steel or aluminum, feature modular components, and include access hatches for medical procedures. For example, at institutions like the San Diego Zoo, transport units have been designed with built-in ports for injections and imaging, minimizing the need for direct physical contact with the animal. These units are designed to maximize safety for both the handlers and the animals while providing functionality for procedures such as x-rays, blood draws, and routine checkups.

Sources:

<https://www.gough.com.au/animal-transport-units.html>

<https://zoo.sandiegozoo.org/animals/komodo-dragon>

One previously identified solution incorporates both collapsibility and segmentation in its design. The windows in this design are made of a combination of metal, plastic, and elastic materials, allowing for adequate ventilation while also providing access points for handling the animal. The segmentation of the boxes allows for ease of transport and adjustability, making it possible to accommodate animals of varying lengths. This is one of the requirements that we are keeping in mind when designing our box because we have to be able to accommodate for Komodo dragons that are 3, 7, and 9 ft long.



Several important design principles were consistent across the systems we studied:

- Segmentation allows the enclosure to be adjusted based on the length of the animal. This is especially important for Komodo dragons, which can range in length from 3 feet to over 9 feet.
- Collapsibility provides ease of storage and transport when the system is not in use.
- Access points and visibility panels allow keepers to perform necessary medical procedures without opening the enclosure, improving both safety and efficiency.
- Durability and safety are maintained through reinforced frames, locking doors, and corrosion-resistant materials.

One cage model uses a combination of metal, plastic, and flexible materials for the window panels. These provide sufficient airflow while also giving staff clear visibility of the animal. The transport unit is long enough to accommodate any animal length and features side access hatches designed to align with common injection or examination zones on the body. The aluminum frame ensures structural integrity during transport.



The current solution used at the Nashville Zoo consists of a large plastic tube. The zookeepers current process requires multiple keepers to transport. However, zookeepers have identified multiple critical flaws in this design. The lack of visibility makes it difficult to monitor the animal, and the absence of access ports significantly complicates medical procedures. Zookeepers report that drawing blood alone can take upwards of 20 minutes, and there is no method for trimming the dragons' nails. Additionally, there are no designated areas for positioning x-ray plates, making imaging procedures nearly impossible. Overall, this design is a makeshift solution that lacks both safety and functionality.

At the Nashville Zoo, the current solution consists of a large cylindrical plastic tube that requires multiple handlers to operate. This design has serious limitations:

- No visibility makes it difficult to monitor the animal during transport or medical procedures.
- No access ports prevent efficient administration of medical care. Drawing blood, for instance, can take over 20 minutes.
- No designated space for x-ray plates means imaging procedures are nearly impossible.

Zookeepers have expressed a strong need for a safer, more accessible system. However, due to limited funding and time, they have not yet developed an improved design.

We aim to improve upon existing systems by combining the most successful features into a single, efficient solution. Our design features three modular units, each 3 feet in length. This approach allows us to adjust the enclosure for dragons of varying sizes.

Key components include:

- A segmented structure for size adaptability.
- Collapsible frames that simplify storage and transport.
- Built-in access hatches for medical procedures, imaging, and basic observation.
- A custom connection system designed to distribute weight evenly and maintain structural stability.

Each segment will feature transparent panels for visibility, glove ports for safe access, and reinforced sidewalls for durability. Medical access panels will be aligned for easy positioning of x-ray plates and safe handling of claws and limbs.

Connection System:

Designing the connection system was one of the most important aspects of the project. The system must:

- Keep the segments aligned and secure during use.
- Support the weight of a Komodo dragon, which can exceed 150 pounds.
- Allow the entire unit to be lifted or rolled without separating.
- Be easy to assemble and disassemble for quick transport or storage.
- Be versatile enough to adapt to changes in the enclosure design.

We explored several options from related fields such as modular crates and equipment transport cases.

Interlocking aluminum frame clamps, toggle latches, and hinge pin systems were among the most relevant designs. These systems are used in applications such as military containers and veterinary equipment and offer proven strength and reliability.

We are currently working on a custom attachment system that uses:

- 3D-printed brackets for precision fitting.
- Stainless steel bolts and hinge locks to ensure alignment and security.
- Internal braces to reinforce the structure across connection points.

One of our goals is for the entire system to be lifted from a central point or supported from the sides without flexing or breaking. This requires exact alignment of the joints and strong materials throughout the frame. Due to our limited budget, we are sourcing parts that are cost-effective while still maintaining the durability needed to safely contain the animal.

Although the exact design may evolve, our current prototype includes a locking rail system with overlapping connectors that ensure each segment stays fully aligned. These components can be locked and unlocked by hand, making setup and breakdown simple for zookeepers in real-world scenarios.



Source: <https://leboeuf rubbishremoval.com/the-garbage-trucks-involved-in-trash-hauling/#:~:text=Complex%20Services%3A%20Front%20Loader%20Recycling%20%26%20Garbage%20Trucks&text=These%20trucks%20use%20a%20hydraulically,the%20top%20of%20the%20truck.>